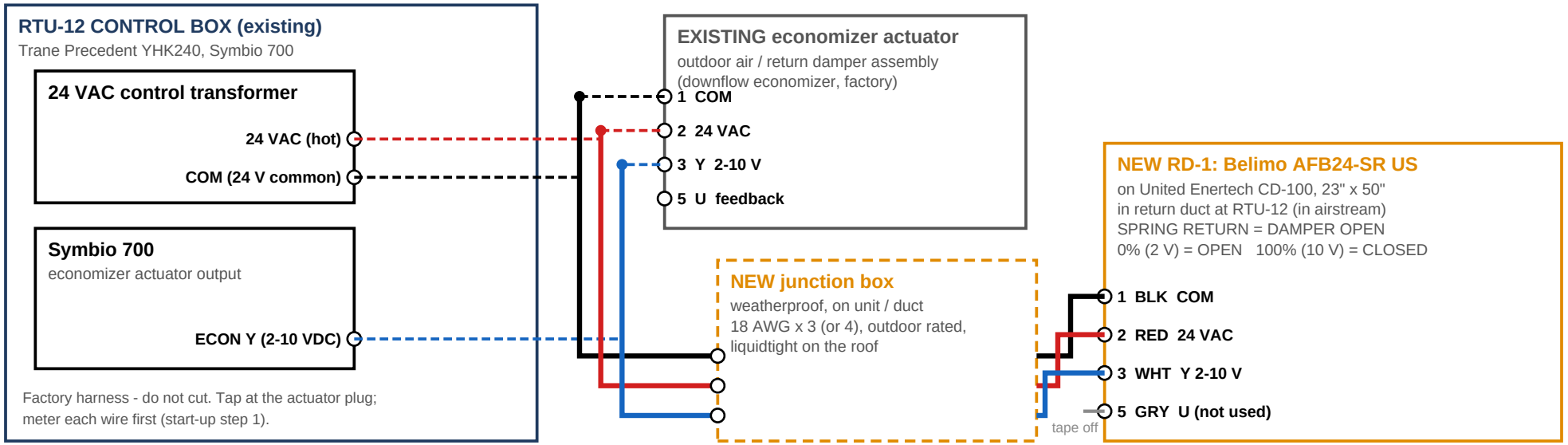
NORTHERN WOLVES
AIR CONDITIONING

RTU-12 RETURN AIR DAMPER (RD-1) - FIELD WIRING

LA Fitness Bronx, NY | Trane Precedent YHK240 (20 ton), Symbio 700, low-leak economizer | Punch list item 3

Northern Wolves AC | 10/17/2026
For installation



How it works

Economizer signal (Y)	Outdoor air damper	RD-1 return damper	Relief shutter
2 V / power off	minimum / closed	OPEN (spring)	closed
min. position (approx. 3-4 V)	minimum OA	nearly open	closed
6 V	about 50 %	about 50 %	starts to open
10 V (full economizer)	100 % open	CLOSED	open, relieving

Start-up check

1. Before tapping: meter the existing actuator - 24 VAC between wires 1-2; 2-10 VDC between 1-3 that changes when the economizer is stepped in Symbio 700 service test.

2. Power off: RD-1 springs fully **OPEN**.

3. Economizer at minimum: RD-1 open (or nearly).

4. Service test 100 % economizer: OA damper open, RD-1 **CLOSED**, relief blades open.

5. Back to normal: RD-1 re-opens smoothly, no binding.

6. Balancer: building pressure at full economizer (target +0.05 in. wg max). If high, set the AFB24-SR end stop so RD-1 does not fully close. Record result in TAB report.

Notes

1. Power RD-1 from the **same** RTU 24 VAC transformer as the economizer actuator and share the **same common** (wire 1). A separate transformer with its own common will make the 2-10 V signal read wrong.

2. Wire 3 (WHT) goes **only** to the economizer 2-10 VDC signal. Never land it on 24 VAC.

3. If the existing actuator turns out to be a communicating type (no plain 2-10 V Y wire), take the signal from its **wire 5 (U, 2-10 V position feedback)** instead of wire 3.

4. AFB24-SR load about 10 VA; check the transformer VA rating has room.

5. Damper: install in the 23" x 50" return duct at the unit, screw frame on all sides and seal the perimeter. Add a 12" x 12" (min.) access door next to the actuator.

6. Old bottom (downflow) return: confirm it is capped and sealed in the curb.

7. Relief shutter (Dayton 1C745 + hood): on the return duct, **building side of RD-1**, as far from the unit as possible, blades horizontal, pivots at top, opening outward. No wiring.

Legend

- 24 VAC hot
- 24 V common
- 2-10 VDC signal (Y)
- existing factory wiring
- new (Northern Wolves)